

Fujitsu FRAM Ferroelectric RAM Product

FRAM is a type of Non Volatile Memory, that use ferroelectric film as a capacitor to store data. This ferroelectric film is polarized by the electric field applied from an external source and remains polarized even with the external electric field removed. With this nature, **FRAM can retain data without data backup battery.**

Advantages

Fast Writing Speed — Overwriting (No Need to Erase), Write Cycle=Read Cycle (No Delay).
100ns speed for both write and read cycles.

High Endurance — Guaranteed 10^{10} Times

Low power consumption — 1/400 or less power consumption compared to EEPROM

High Tamper Resistance — Data written in FRAM can not be stolen by physical analysis

Ecology Memory — Operating like SRAMs, with no requirement for battery backup, less 65% CO₂ during process manufacturing **FRAM does not need polling sequence after writing to memory such as case of Flash memory or EEPROM**

Comparison of FRAM with another memory devise

	FRAM	EEPROM	FLASH	SRAM
Type	Non Volatile	Non Volatile	Non Volatile	Volatile
Method writing	Overwriting	Erase(byte)+Write	Erase(sector)+Write	Overwriting
Write Cycle Time	150ns	3ms	1s	55ns
Endurance	10billion	1 million	1 million	Unlimited

Target Application

- ◆ 3-Phase power meter
- ◆ Automatic Meter Reading (AMR)
- ◆ POS machine, ATM
- ◆ Industrial meters
- ◆ Industrial process control
- ◆ OA machine



**FRAM is available with Fujitsu distributors
who can sell in local Indian rupee & maintain sufficient stock**

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